Eashan Patel

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EDUCATION

Rutgers University Honors College - B.S. in Computer Science, B.A. in Data Science **September 2022 - May 2026**

- Relevant Coursework: Data Structures, Computer Architecture, Design of Algorithms, Intro to Artificial Intelligence, Brain-Inspired Computing, Machine Learning, Regression Methods, Applied Statistics, Statistical Inference for Data Science
- GPA: 3.9

WORK EXPERIENCE

Software Engineer Intern **June 2024 – August 2024**

Voya Financial Remote

- Authored technical documentation for an **extJS codebase** maintaining client financial portals
- Developed flowcharts and Excel tracking systems to support the team's migration to a **new Salesforce cloud**
- Implemented **Java code** to track user interactions within retirement financial portals
- Collaborated with developers, analysts and stakeholders in an **Agile Scrum** environment

Online Instructor **June 2023 – August 2023**

iDTech Remote

- Taught **Python, Java, JavaScript and React Native** to **50+ students** in both group and one-on-one virtual classroom settings
- Collaborated with instructors and management to improve curriculum structure and student engagement
- Designed personalized lesson plans and adaptive teaching strategies to meet individual student needs

PROJECTS

ScarletParking | *JavaScript, Expo, React Native*

- Mobile application allowing Rutgers students to select parking passes and view valid parking locations via map and list views
- Shipped interactive map functionality and real-time filtering for improved usability
- Reached **Top 100** in Apple's Navigation category with **12,000+** impressions, **1,500+** downloads, and a **5-star** rating

RU Subletting | *TypeScript, Next.js, React, Tailwind, Supabase, PostgreSQL, REST APIs*

- Developing a full-stack web application where college students browse and post sublease listings through a marketplace
- Building the frontend with **Next.js, React and TypeScript**, including filtering by price, dates and amenities, detailed listing pages, and saved listings
- Designing a **PostgreSQL** schema on Supabase with **row-level security policies** for access control, integrating its **REST API** and writing custom server-side routes for server-only operations

Wrapped Anytime | *JavaScript, React, Bootstrap*

- Web application that utilizes the **Spotify-API** to access user data and display top artists and songs over different time periods
- Implemented **OAuth-based** user authentication, access token handling, and dynamic **API-driven** content rendering

Robust Keyword Detection Using Spiking Neural Networks | *Python, PyTorch, snnTorch and Speech2Spikes*

- Created a **Spiking Neural Network (SNN)** using multi-layer **Leaky Integrate-and-Fire (LIF)** neurons with surrogate gradient training
- Achieved **96% accuracy** on clean speech for binary keyword classification
- Designed experiments using noise distortions and demonstrated improved generalization, reaching **93.7% accuracy** on distorted audio while maintaining **>94%** clean-speech performance

Machine Learning Madness | *Python, XGBoost, scikit-learn, pandas, NetworkX*

- Built an **end-to-end pipeline** using Kaggle NCAA data to predict tournament point differential and win probability for men's and women's brackets, engineering 200+ matchup features from box scores, seeds and other rankings
- Adapted **Google's PageRank** into a margin-of-defeat-weighted team-strength metric ($R^2 = 0.335$ vs. point differential) that ranked among the top **XGBoost** features
- Found that combining team rankings, PageRank-based strength rating and box-score stats predicted tournament games more accurately than seed alone, cutting average prediction error to **8.9 points** for men's games (from 9.2) and **10.3** for women's (from 11.3)

SKILLS

- **Languages:** Python, JavaScript, Java, TypeScript, C, SQL, HTML, CSS, ExtJS
- **Frameworks & Tools:** React, React Native, PostgreSQL, Next.js, Expo, Node.js, Bootstrap, Tailwind, PyTorch, snnTorch, XGBoost, scikit-learn, pandas, NumPy, REST APIs, NetworkX, Matplotlib, Git/GitHub, Excel, Cursor, Claude Code
- **Core Development Skills:** Relational Databases, Data Structures & Algorithms, Machine Learning, Neural Networks, Statistical Modeling, REST API Integration, OAuth Authentication, Full Stack Development, Mobile App Development, AI-Assisted Development, Technical Documentation, Agile/Scrum